

Different ages

A company stores the age and salaries of its employees.

Write a program that gives us how many different aged people are employed.



Input

The first line of the standard input contains the count of employees ($1 \leq N \leq 100$). The next N lines contain the age ($1 \leq A \leq 100$) and salary ($1 \leq S \leq 2\,000\,000$) of an employee each.

Output

The first line of the standard output should contain the count of different aged employees.

Example

Input

```
6
42 150000
33 100000
25 600000
66 80000
25 200000
42 800000
```

Output

```
4
```